

Chapter 10: The Tools of Quality

ISE 530 – Quality in Supply Chain Management
KFUPM

Learning Objectives

- Apply **process maps and extended value stream** maps to improve supply chain performance.
- Use Ishikawa's **seven basic quality tools (B7)** to identify and solve SCM problems.
- Apply the **new seven management tools (N7)** for strategic decision-making in supply chains.
- Design and interpret dashboards, balanced scorecards, and spider charts for SCM.
- Conduct **hands-on exercises and simulations** to reinforce each tool in a practical SCM context.

Opening Case: The Boeing 787 Dreamliner Supply Chain Crisis

What Happened?

- Boeing outsourced ~70% of 787 components to 50+ suppliers across 135 sites worldwide.
- Poor quality from Tier 2 & 3 suppliers led to 3+ years of delays.
- Fuselage sections arrived with thousands of defects; fasteners didn't fit.
- **Estimated cost overrun**: \$12–18 billion.

Root Causes (Quality Perspective)

- No **standardized process maps** across the supply chain.
- **Lack of control charts and SPC at** sub-tier suppliers.
- **No Pareto analysis to** prioritize the most critical defect categories.
- **Missing cause-and-effect analysis** for systemic failures.
- **If Boeing had used these quality tools effectively, would the outcome differ?**

DISCUSSION

Think – Pair – Share: Your Supply Chain Experience

- Recall a product you purchased that had a quality defect. What was the defect? **Where in the supply chain** do you think it originated?
- Discuss which **quality tool(s)** could have prevented the defect.
- How many of **these quality problems** were caused by the *manufacturer vs. a supplier?*

The Quality System in Supply Chain Context

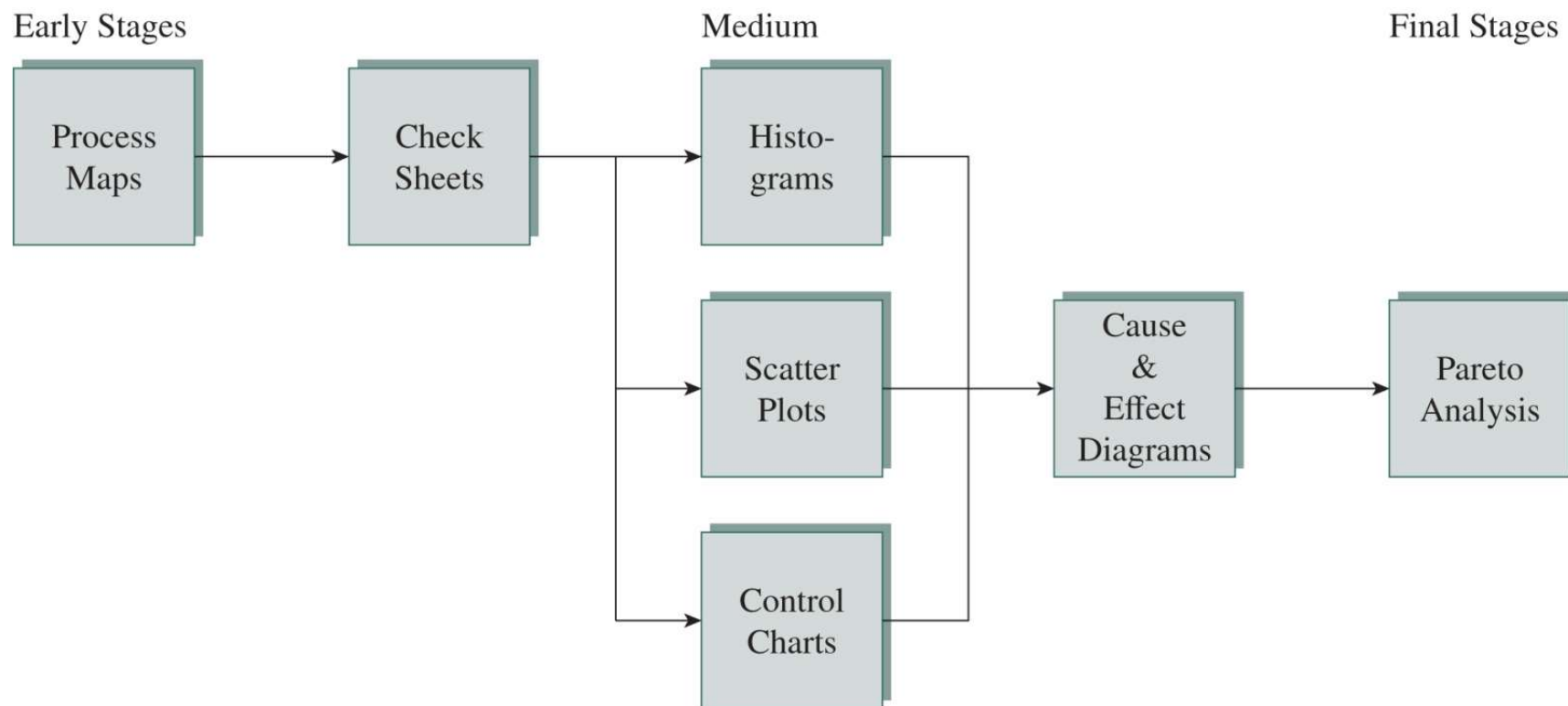
- A quality system balances: Supply Chain, Marketing, Accounting, HR, Operations, Engineering, Strategy.
- In SCM, quality must extend **beyond the factory walls to every tier** of the supply network.
- The quality system is a LIVING system, not a one-time project but a **continuous cycle**.
- Quality failures at any node of the supply chain cascade downstream to the final customer.



Part I: The Basic Seven (B7) Tools

Ishikawa's Foundation for Data-Driven Quality Improvement

Overview: Ishikawa's Basic Seven (B7) Tools

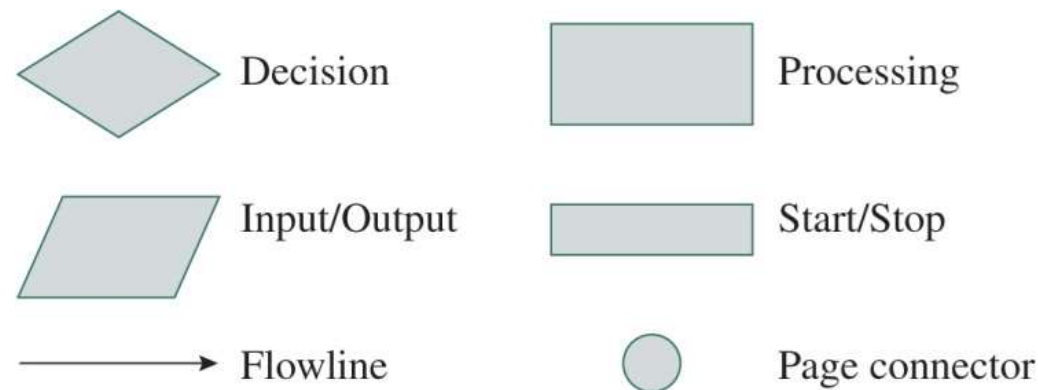


Tool 1: Process Maps

Visualizing and improving supply chain processes

Process Maps: Key Concepts

- A picture of a process as it currently exists, using standard symbols:



- Simple rules: Chart the process start to finish; all arcs leave/enter a symbol.
- Develop a general map first, then add detail (sub-flowcharts) for each element.
- Walk the process – interview people as they perform the work.
- Identify value-adding vs. non-value-adding steps.
- Before simplifying: ask whether the work needs to be done at all.

SIPOC Diagram: Extending Process Maps for SCM

SIPOC = Supplier → Inputs → Process → Outputs → Customers

- Particularly useful when:
 - It's not clear who the internal/external customers are.
 - Specifications for inputs are undefined or inconsistent.
 - Customer requirements need clarification across the supply chain.

In SCM: Each link in the chain has its own SIPOC, and they must align end-to-end.

- *A SIPOC diagram is the starting point for any process improvement project.*

Scenario

- Amazon's fulfillment centers process millions of orders daily.

The 'Receive → Stow → Pick → Pack → Ship' process is mapped in extreme detail.

- Every step is timed; non-value-adding steps are eliminated.
- Kiva robots were introduced after process mapping revealed excessive walking time (60–70% of picker's time).

Quality Impact

- Order accuracy improved from 99.5% to 99.97%.
- Cycle time reduced by 75% with robotic stow/pick.
- SIPOC used to define interfaces between warehouse zones.
- Lesson: You cannot improve what you cannot visualize.

EXERCISE

Hands-On Exercise: Map a Coffee Shop Supply Chain

Instructions

- You are the quality team for a local coffee chain (e.g., KBS's Café).
- Map the supply chain process from coffee bean sourcing to serving a cup.
- Supplier → Roasting → Warehouse → Store → Barista → Customer.
- Use standard process map symbols on the provided flip chart.
- Identify at least 3 non-value-adding steps.

Deliverables

- A complete SIPOC diagram for the chain.
- A process map of the 'current state'.
- Mark NVA (non-value-adding) steps with a red X.
- Propose an 'improved state' process map.

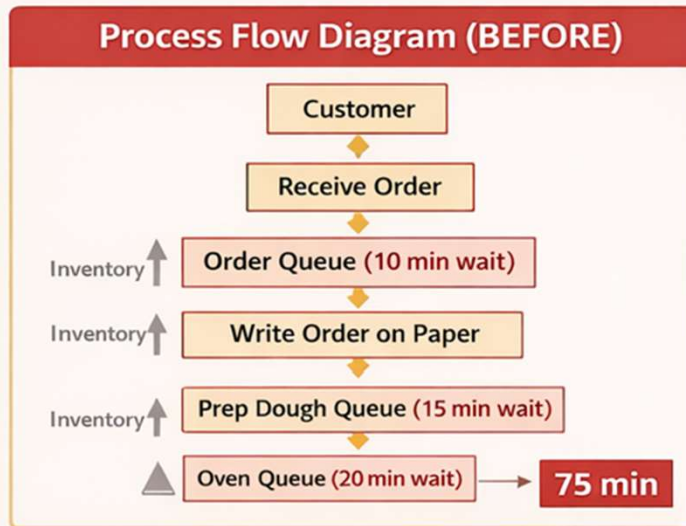
Discussion: Which NVA steps are hardest to eliminate? Why?

Value Stream Mapping (VSM) in Supply Chains

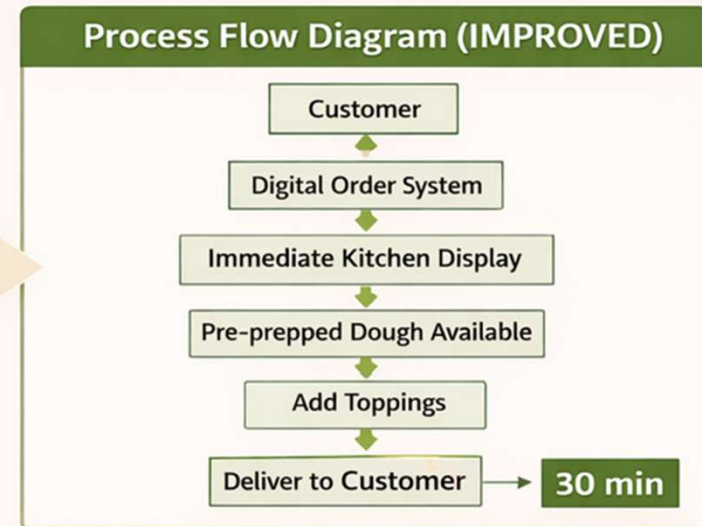
- Extended VSM **goes beyond the factory** to map the entire supply chain.
- Maps both **material flow AND information flow** between supply chain partners.
- **Key metrics captured:** Lead time, Processing time, Inventory levels, Transport time.
- Current-State Map → Identify waste → Future-State Map → Implementation plan
- **Collaboration** between customer and suppliers reduced total lead time significantly.
- VSM reveals the 'hidden factory', rework, waiting, and excess inventory between tiers.



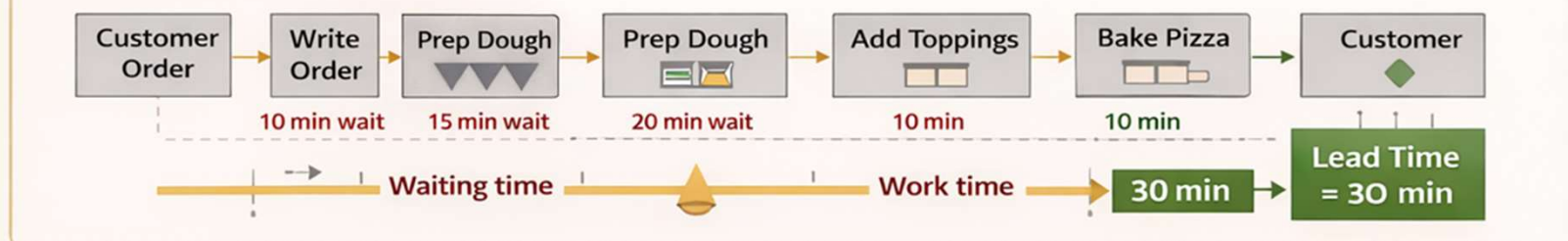
Pizza Shop Example: Process Flow vs. Value Stream Map



VS.



Easy Value Stream Map



Tool 2: Check Sheets

Systematic data collection at the source

Check Sheets: Collecting Quality Data in SCM

- A structured form for collecting and organizing data – the foundation for all analysis.
- Tabular (tally marks), Location-based (defect maps), Computer-based (digital forms).

Setup steps:

- 1. Identify common defects/issues in the process.
- 2. Create a table: defect types in rows, time periods in columns.
- 3. Workers mark occurrences as they happen – real-time data capture.

In SCM: Check sheets are used at receiving docks, during in-process inspection, and at shipping verification points.

Modern evolution: Digital check sheets on tablets, barcode/QR scanning, IoT sensors.

Case Study: Toyota's Supplier Quality Check Sheets

Background

- Toyota requires all Tier 1 suppliers to use standardized check sheets.
- At the receiving dock, inspectors use defect location diagrams (schematic check sheets).
- Each defect is categorized: dimensional, surface, functional, packaging.
- Data feeds directly into Toyota's Supplier Quality Database.

Results & Lessons

- Suppliers with check sheet compliance: 40% fewer returns.
- Defect trends visible within days, not months.
- Check sheets led to discovery that 60% of surface defects originated during transport, not manufacturing.
- Action: Packaging redesign reduced surface defects by 72%.
- Lesson: The simplest tool often yields the biggest insights.

Example 10-2

- **Problem:** A copying company wants to set up a check sheet so that it can keep track of error sources.
- Following are the major error types with frequencies:

Type of Problem	Frequency	Percentage
Setup routines are not standardized	315	52.1%
Equipment needed for setup is missing	124	20.5
Internal and external setup tasks are not separated	87	14.4
Extensive machine resetting and paper change are needed	56	9.2
Other	<u>23</u>	<u>3.8</u>
Total	605	100%

Example 10-2

- **Solution:** Figure 10-8 shows a check sheet for these data. The check sheet will be kept to monitor how well workers are adhering to the new procedures.

Problem Type	Monday	Tuesday	Wednesday	Thursday	Friday	Total
Setup routines not standardized	3	2	4	3	2	14
Missing equipment for setup	1	3	2	2	1	9
Failure to separate internal and external tasks	2	1	3	2	2	10
Extensive machine resetting and paper change	4	3	5	4	3	19
Other	1	0	1	1	0	3
Daily Total	11	9	15	12	8	55

Figure 10-8

Tool 3: Histograms

Understanding the distribution and shape of process data

Histograms

- Graphical representations of data in bar format
- Used for continuous numerical data and to observe the shape of the data

Rules for Developing Histograms

- The width of the histogram bars must be consistent.
- The classes must be mutually exclusive and all-inclusive (or collective exhaustive).
- A good rule of thumb for the number of classes is $2^k \geq n$.
 - Where n is the number of raw data values and k is the number of classes
 - Solving for k , we obtain $k \geq \log n / \log 2$, therefore:

Number of Observations	Number of Classes
9 to 16	4
17 to 32	5
33 to 64	6
65 to 128	7
129 to 256	8

Example 10-3

- **Problem:** The Big City Cafeteria wants to determine the distribution of its sales during lunchtime. On a given day, the manager randomly selects 40 sales from the sales register receipt. Develop a histogram of the sales.
- The following table shows the sales (in dollars).

4.51	0.79	4.19	2.29
5.96	3.49	2.25	3.45
2.24	5.25	5.36	1.15
7.28	5.25	4.29	5.25
3.96	6.79	4.66	3.56
8.22	2.56	5.25	3.33
5.55	2.24	8.95	2.49
5.25	2.26	0.79	5.25
4.11	6.11	5.25	4.56
1.15	5.25	2.21	5.25

Example 10-3

- **Solution:** It is helpful to compute the mean, standard deviation, maximum value, and minimum value when developing a histogram because the histogram is often used to determine whether the data are normally distributed.
- **Following are these statistics from the previously given data:**
 - Mean = 4.20
 - Maximum value = 8.95
 - Minimum value = .790
 - Difference = 8.16
 - Sum = 168

Example 10-3

- $k \geq \log 40 / \log 2$
- $k \geq 5.32$
- The number of classes is 6, therefore:
 - Class width = $8.16 / 6 = 1.36 \sim 1.40$
 - Classes: 0.76-2.15, 2.16-3.55, 3.56-4.95, 4.96-6.35, 6.36-7.75, 7.76-9.15

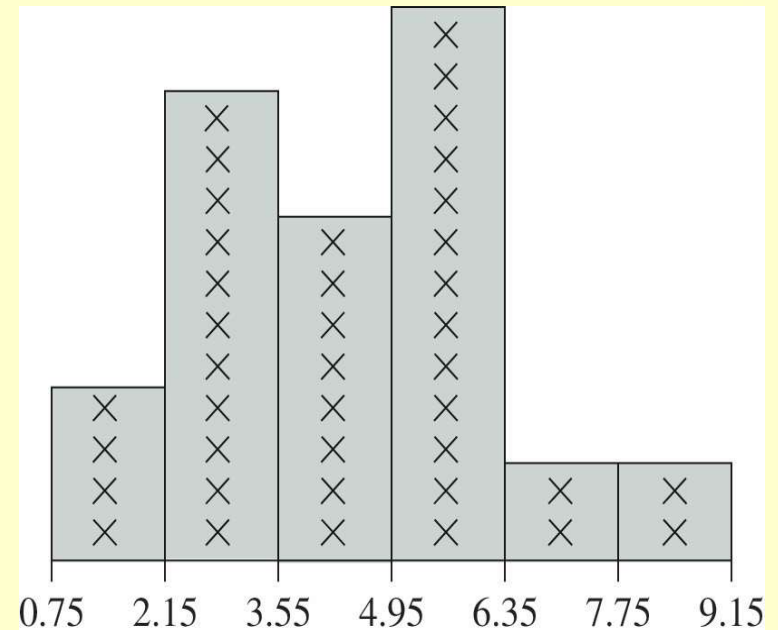


Figure 10-9

Tool 4: Scatter Diagrams

Discovering relationships between quality variables

Scatter Diagrams

- Charts used to examine the relationships between variables; also called *scatter plots*.
- Sometimes used to identify indicator variables in organizations.
- **Steps to set up a scatter plot:**
 1. Determine your x (independent) and y (dependent) variables.
 2. Gather process data relating to the variables identified in Step 1.
 3. Plot the data on a two-dimensional plane.
 4. Observe the plotted data to see whether there is a relationship between the variables.

Scatter Diagrams

Scatter plot of the relationship between conformance data and prevention and appraisal quality-related costs in a real firm

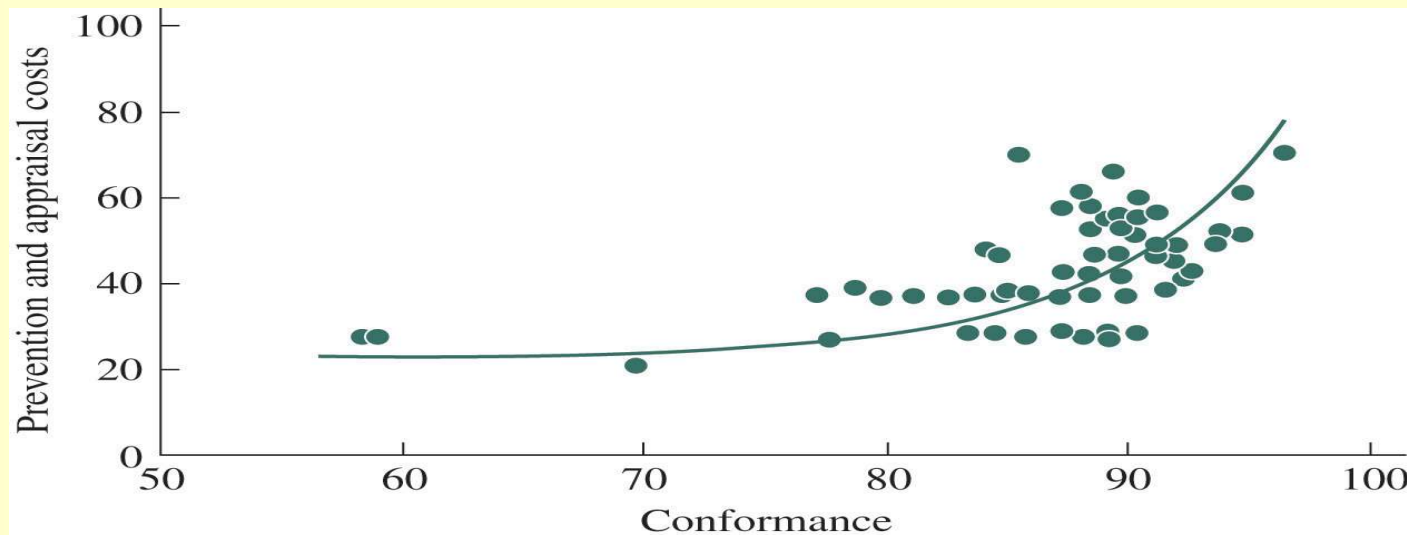


Figure 10-10

Example 10-4

- **Problem:** Healthy People, Inc., a company specializing in home healthcare solutions for U.S. consumers, was a growing company. The company wanted to study the relationship between absenteeism and the number of overtime hours worked by employees. Thirty employees were randomly selected, and numbers of overtime hours were graphed against numbers of days absent for the previous year. (See Figure 10-11.)

Example 10-4

Employee	Hours of Overtime	Days Absent
1	243	3
2	126	2
3	86	0
4	424	6
5	236	3
6	128	0
7	0	0
8	126	2
9	324	3
10	118	0
11	62	0
12	128	3
13	460	6
14	135	1
15	118	1

Employee	Hours of Overtime	Days Absent
16	260	2
17	0	1
18	126	1
19	234	2
20	246	3
21	120	1
22	80	0
23	112	1
24	237	3
25	129	2
26	24	1
27	36	0
28	128	2
29	246	3
30	326	6

Example 10-4

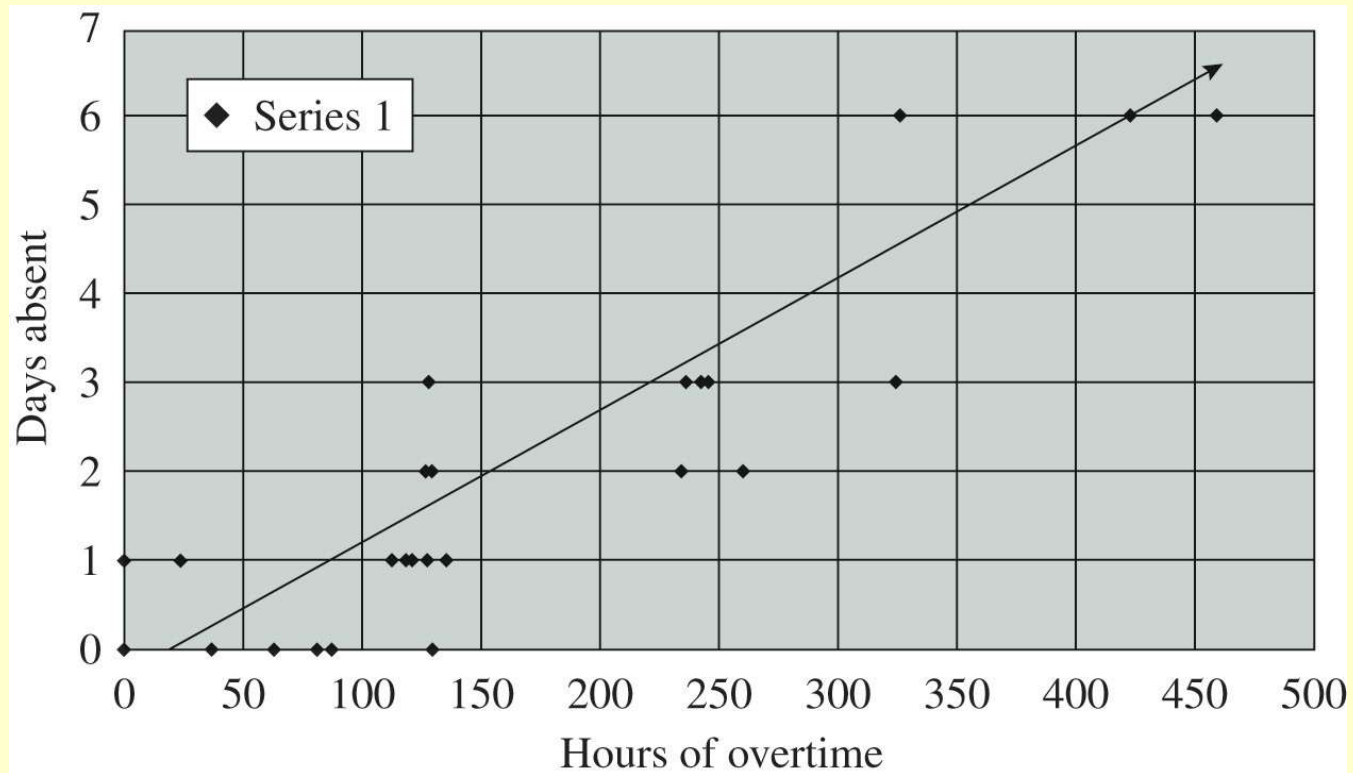


Figure 10-11

Case Study: Does Supplier Distance Affect Quality?

Scenario

- A medical device manufacturer sources components from 15 suppliers worldwide.
- Quality team hypothesized: longer shipping distances → more transit damage.
- Data collected: Distance (km) vs. Incoming defect rate (%) for 15 suppliers.
- Scatter plot showed moderate positive correlation ($r = 0.62$).

Analysis & Action

- Closer inspection: 3 outlier suppliers (very far, but low defect rate) used custom shock-absorbing packaging.
- Root cause was not distance itself, but packaging quality.
- Action: Standardized packaging specs for all distant suppliers.
- Result: Defect rate for distant suppliers dropped from 4.2% to 1.1%.
- Lesson: Scatter plots reveal correlations, but root cause analysis reveals solutions.

Tool 5: Control Charts

Monitoring process stability across the supply chain

Control Charts: The Voice of the Process

- Charts used to determine whether a process produces consistent, predictable output.
- Structure: Center Line (CL = mean), Upper Control Limit (UCL), Lower Control Limit (LCL).
- UCL and LCL are typically set at ± 3 standard deviations from the mean.
- A process 'in control' shows only common cause (random) variation.
- A process 'out of control' shows special cause variation – needs investigation.
- Out-of-control signals: Point beyond limits, 7+ points on one side, trends, cycles.
- SCM applications: Monitor supplier delivery times, incoming defect rates,
 - order accuracy, fill rates, and any measurable quality characteristic.

Case Study: SPC at a Pharmaceutical Supply Chain

Background

- A pharmaceutical company monitors tablet weight from 3 suppliers.
- Target weight: 500 mg $\pm$ 5 mg (Spec: 495–505 mg).
- Supplier A: $\bar{x}$ = 500.1, R = 3.2 $\rightarrow$ Stable, capable (Cpk = 1.56).
- Supplier B: $\bar{x}$ = 499.5, R = 6.8 $\rightarrow$ Stable, but marginal (Cpk = 0.98).
- Supplier C: $\bar{x}$ = 501.2, R = 4.1 $\rightarrow$ Out of control (shift detected).

Actions Taken

- Supplier A: Continue monitoring; gold-standard supplier.
- Supplier B: Capability improvement project initiated; reduce variation.
- Supplier C: Production stopped; root cause = worn die tooling.
- After fix: Supplier C Cpk improved from 0.72 to 1.45.
- Lesson: Control charts catch problems before they reach the customer.
- Without SPC: defective tablets would have shipped to pharmacies.

Simulation: The Red Bead Experiment (Deming)

Setup (30 minutes)

- Materials: Bowl with 800 white beads + 200 red beads, a paddle with 50 holes.
- 6 'workers' take turns scooping beads (= producing output).
- Red beads = defects. White beads = good output.
- A 'supervisor' records the number of red beads per scoop.
- Run 4 rounds per worker (24 data points).

Learning Objectives

- Plot the results on a control chart (p-chart).
- Are any workers 'out of control'?
- Key insight: Variation is inherent in the SYSTEM, not caused by individual workers.
- Discussion: Should we fire the worker with the most red beads?
- Deming's lesson: Management owns the system; workers work in the system.
- Application: Blaming a supplier for systemic variation is counterproductive.

Tool 6: Cause-and-Effect (Fishbone) Diagrams

Systematic root cause analysis

Cause-and-Effect (Ishikawa Diagrams)

- A tool to help move to lower levels of abstraction in solving problems
- Looks like the skeleton of a fish
 - Problem = Head
 - Ribs = Major causes
 - Bones = Subcases

Cause-and-Effect (Ishikawa Diagrams)

Steps to create a cause-and-effect diagram:

- State the problem clearly in the head of the fish.
- Draw the backbone and the ribs by asking participants to identify major causes of the problems labeled in the head of the diagram.
- Continue to fill out the diagram asking “Why?” about each problem or cause of a problem until the fish is filled out.
- View the diagram and identify core causes.
- Set goals to address the core causes.

Cause-and-Effect Diagram: Wobbling Saw Blade

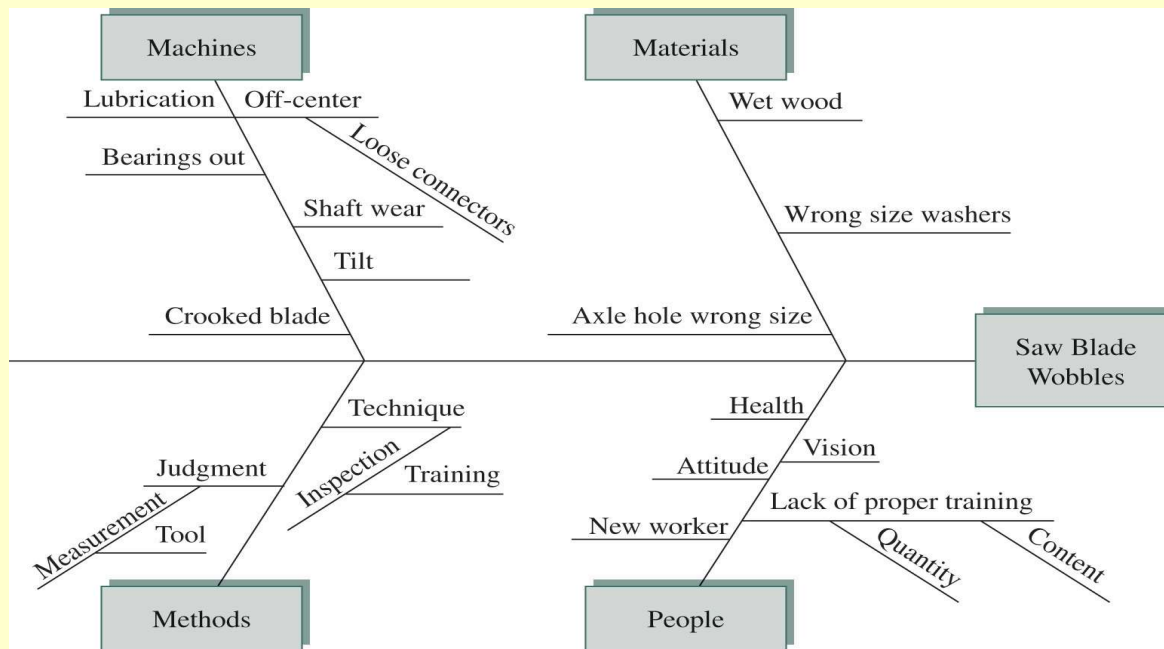


Figure 10-13

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Case Study: Samsung Galaxy Note 7 Battery Failures

The Problem

- In 2016, Samsung Galaxy Note 7 phones caught fire worldwide.
- Samsung recalled 2.5 million units – cost: \$5.3 billion.
- Root cause: Battery defects from TWO different suppliers.
- Supplier A (Samsung SDI): Electrode deflection in upper corner.
- Supplier B (ATL): Missing insulation tape + welding burrs.

Fishbone Analysis

- Material: Battery cell design too tight for the casing.
- Supplier: Both suppliers had different but critical defects.
- Method: Insufficient incoming inspection protocols.
- Measurement: Battery safety tests did not catch the failure mode.
- Management: Pressure to launch before Apple iPhone 7.
- Lesson: A fishbone diagram spanning the full supply chain would have identified the multi-supplier risk earlier.

Tool 7: Pareto Charts

Prioritizing the vital few from the trivial many

Pareto Charts

- **Rules for constructing Pareto charts:**
 - Information must be selected based on types or classifications of defects that occur as a result of a process.
 - Data must be collected and classified into categories.
 - A frequency chart must be constructed, showing the number of occurrences in descending order.
- **Steps in Pareto analysis:**
 1. Gather categorical data relating to quality problems.
 2. Draw a frequency chart of the data.
 3. Focus on the tallest bars in the frequency chart first when solving the problem.

Example 10-6

- **Problem:** A copying company is concerned because it is taking too long for operators to set up new printing jobs. They decide to use Pareto analysis to find out why setup times are taking so long.
- The data gathered reflect the following major causes:

Type of Problem	Frequency	Percentage
Setup routines are not standardized	315	52.1%
Equipment needed for setup is missing	124	20.5
Internal and external setup tasks are not separated	87	14.4
Extensive machine resetting and paper change are needed	56	9.2
Other	23	3.8
Total	605	100%

Example 10-6

- **Solution:** First, order the problems by frequency and compute the percentage of problems related to each cause, and then draw a frequency chart of the results.

Type of Problem	Frequency	Percentage
Setup routines are not standardized	315	52.1%
Equipment needed for setup is missing	124	20.5
Internal and external setup tasks are not separated	87	14.4
Extensive machine resetting and paper change are needed	56	9.2
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Example 10-6

Pareto Chart

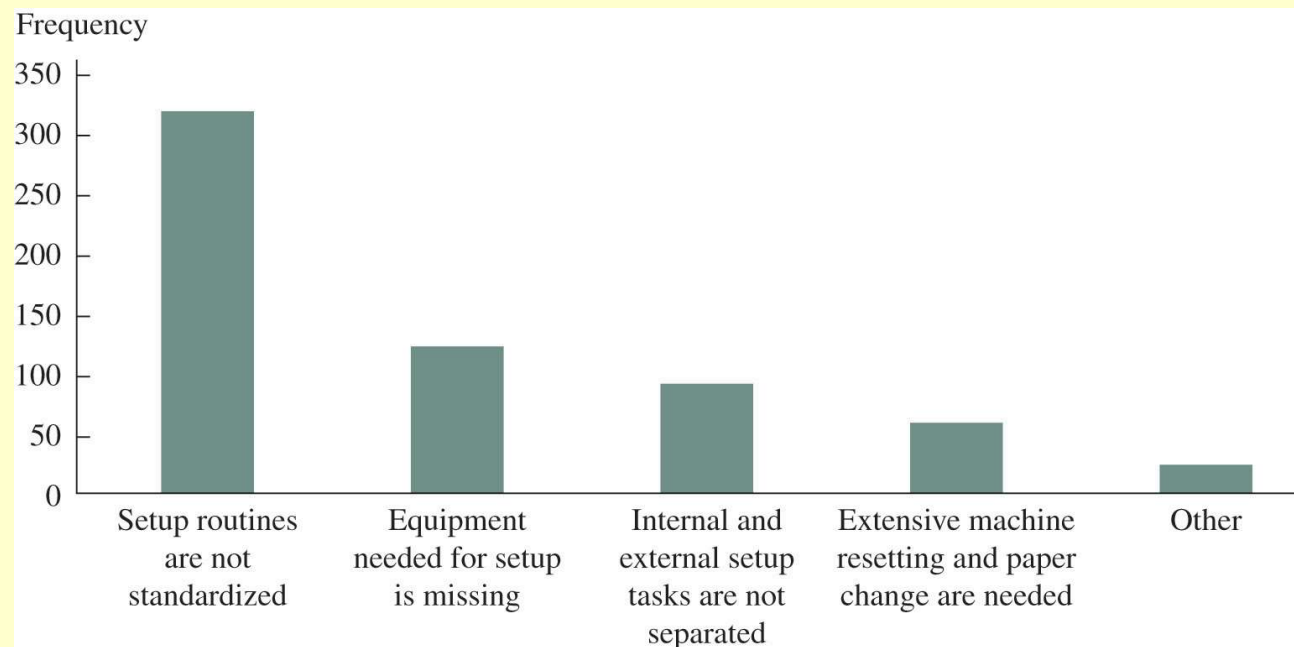


Figure 10-15



Part II: The New Seven (N7) Management Tools

Strategic Tools for Managerial Decision-Making in SCM

Overview: The New Seven (N7) Tools

- Designed for MANAGERIAL problem-solving (vs. B7 for operational/data problems).
- 1. Affinity Diagram – Organize brainstorming ideas into themes.
- 2. Interrelationship Digraph – Identify causal relationships between issues.
- 3. Tree Diagram – Break down goals into actionable tasks.
- 4. Prioritization Matrix – Make decisions based on multiple criteria.
- 5. Matrix Diagram – Show relationships between multiple dimensions.
- 6. Process Decision Program Chart (PDPC) – Plan for contingencies.
- 7. Activity Network Diagram – Plan and control project timelines (PERT/CPM).
-
- These tools are used sequentially: Affinity → Digraph → Tree → Prioritization → Matrix → PDPC → Network.

Affinity Diagram: Organizing Ideas

- Helps a group converge on themes from a brainstorming session.
- Steps:
 - 1. State the problem clearly.
 - 2. Each team member writes ideas on sticky notes (one idea per note, 10 min).
 - 3. Place all notes on a large surface (wall/table).
 - 4. SILENTLY sort notes into groups with similar themes.
 - 5. If you disagree with a placement, move it silently – no talking!
 - 6. Consensus is reached when cards stop moving.
 - 7. Label each group with a header card.
- Best for: Unstructured problems, capturing diverse perspectives, starting a project.

Exercise: Affinity Diagram – Improving Supplier Relationships

Instructions (20 min)

- Problem: ‘What are the key issues affecting our supplier relationships?’
- Each person: Write 5–8 ideas on sticky notes (3 min).
- Place all notes on the table.
- Sort silently into groups (5 min). NO TALKING during sorting!
- Label each group with a header.
- Expected themes: Communication, Quality, Cost, Delivery, Trust, Technology.

Debrief Questions

- How many theme groups did your team identify?
- Were there any surprises – ideas you hadn’t considered?
- Which theme has the most sticky notes? What does that suggest?
- How does silent sorting prevent ‘groupthink’?
- How would you use this output in a real improvement project?

Interrelationship Digraph: Finding Root Drivers

- Shows causal relationships between the themes identified in the affinity diagram.
- Steps:
 - 1. Take the header cards from your affinity diagram.
 - 2. For each pair of themes, ask: 'Does A cause or influence B, or vice versa?'
 - 3. Draw an arrow from cause → effect.
 - 4. Count incoming arrows (IN) and outgoing arrows (OUT) for each theme.
 - 5. Themes with the most OUT arrows are ROOT DRIVERS (causes).
 - 6. Themes with the most IN arrows are KEY OUTCOMES (effects).
 - 7. Focus improvement efforts on the root drivers.
- In SCM: This tool reveals which supplier issues are causing other issues downstream.

EXERCISE

Exercise: Interrelationship Digraph – Continue from Affinity Diagram

- Using the theme headers from your affinity diagram exercise, build an interrelationship digraph.
- For each pair of themes, discuss and draw an arrow from cause to effect.
- Example: Does 'Poor Communication' cause 'Quality Issues'? Or the reverse? Or both?
- Count IN and OUT arrows for each theme. Record in a table:

Theme	OUT arrows	IN arrows	Net (OUT – IN)	Driver or Outcome?
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- Identify the top 2 root drivers. These should be the priority for improvement.
- Present your digraph and explain your reasoning (3 min per team).
- Discussion: Did the root driver surprise you? How does this change your action plan?
- Time: 15 minutes.

Tree Diagram: Breaking Down Goals into Actions

- Takes a high-level goal and breaks it into increasingly detailed, actionable tasks.
- Structure: Goal → Major Tasks → Sub-Tasks → Specific Actions.
- Steps:
 1. Choose the most important issue from your digraph.
 2. State it as a goal: 'Improve supplier communication.'
 3. Ask: 'What major tasks are needed?' → Level 1 branches.
 4. For each branch, ask: 'What specific steps are needed?' → Level 2.
 5. Continue until actionable items are reached.
- In SCM: Tree diagrams are used to decompose quality improvement goals, new supplier onboarding processes, and corrective action plans.

Prioritization Matrix: Data-Driven Decision Making

- Used to evaluate and rank alternatives based on multiple weighted criteria.
- Steps:
 - 1. Define the goal, alternatives, and evaluation criteria.
 - 2. Weight each criterion (sum of weights = 1.0).
 - 3. Rate each alternative against each criterion.
 - 4. Multiply ratings by weights and sum for each alternative.
 - 5. Rank alternatives by total weighted score.
- SCM Example: Selecting the best supplier using criteria like:
 - Quality (0.30), Cost (0.25), Delivery (0.20), Flexibility (0.15), Innovation (0.10).
- Removes subjectivity from supplier selection and other strategic decisions.

Case Study: Supplier Selection Using Prioritization Matrix

Scenario

- An automotive OEM must select 1 of 3 suppliers for a critical brake component.
- Criteria and weights determined by cross-functional team:
- Quality (0.35), Cost (0.20), Delivery (0.20),
- Technical Capability (0.15), Financial Stability (0.10).
- Supplier ratings (1–5 scale) collected from audits and performance data.

Results

- Supplier X: Quality=5, Cost=3, Delivery=4, Tech=4, Finance=5
→ Score: 4.25
- Supplier Y: Quality=3, Cost=5, Delivery=3, Tech=3, Finance=4
→ Score: 3.45
- Supplier Z: Quality=4, Cost=4, Delivery=5, Tech=5, Finance=3
→ Score: 4.30
- Decision: Supplier Z wins (barely over X).
- Sensitivity analysis: If Quality weight increased to 0.40, Supplier X wins.
- Lesson: Transparent criteria prevent 'lowest cost' bias in supplier selection.

Exercise: Supplier Selection Prioritization Matrix

Scenario

- Your company needs a new logistics provider for Middle East distribution.
- 3 candidates: DHL, Aramex, and a local provider (FastShip).
- As a team, define 5 evaluation criteria and assign weights (must sum to 1.0).
- Rate each provider 1–5 on each criterion.
- Compute weighted scores and rank.

Discussion

- Did all teams choose the same criteria and weights?
- How sensitive is the result to weight changes?
- What happens if you change the most important criterion?
- How would you validate your ratings with real data?
- Should you single-source or dual-source? Why?
- Time: 20 minutes.

Matrix Diagram: Multi-Dimensional Relationships

- A brainstorming tool showing relationships between two or more sets of items.
- Types: L-shaped (2 dimensions), T-shaped (3), Y-shaped (3), X-shaped (4), Roof-shaped.
- Symbols indicate strength: ● Strong, ○ Moderate, △ Weak.
- SCM application: Relate customer requirements to supplier capabilities (like QFD House of Quality).
- Example: Map 'Customer needs' vs. 'Supplier capabilities' to identify gaps.
- Steps: (1) Determine dimensions, (2) Choose matrix type, (3) Place relationship symbols.

Process Decision Program Chart (PDPC)

- A contingency planning tool – ‘What could go wrong, and how do we respond?’
- Built on top of the tree diagram by adding:
 - Level 3: ‘What unexpected things could happen?’
 - Level 4: ‘What countermeasures can we take?’
- Mark feasible countermeasures with ○ and infeasible with X.
- In SCM: Essential for supply chain risk management.
 - – What if a key supplier has a factory fire? → Countermeasure: Dual sourcing.
 - – What if a port is blocked? → Countermeasure: Alternative routing.
 - – What if demand spikes 200%? → Countermeasure: Safety stock, flexible contracts.
- PDPC is the supply chain manager’s risk playbook.

Activity Network Diagram (PERT/CPM)

- A project management tool for planning, scheduling, and controlling projects.
- Also known as PERT (Program Evaluation and Review Technique) or Critical Path Method (CPM).
- Key concepts:
 - – Nodes represent activities; arcs show dependencies.
 - – Critical Path = longest path through the network = minimum project duration.
 - – Slack = how much an activity can be delayed without delaying the project.
- SCM application: New product introduction, supplier onboarding, quality improvement projects.
- Essential for managing complex supply chain transformation projects.

Exercise: PDPC for Supply Chain Risk Management

Scenario

- You supply automotive parts to a major OEM.
- Your supply chain: Raw material supplier (Turkey) → Your factory (Saudi Arabia)
- → Assembly plant (Germany) → Dealer network (Europe).
- Create a PDPC for the goal: 'Ensure uninterrupted supply to the OEM.'

Tasks

- 1. Build a tree diagram with 3 major task branches.
- 2. For each branch, identify 2–3 things that could go wrong.
- 3. For each risk, brainstorm 1–2 countermeasures.
- 4. Mark each countermeasure as feasible (o) or infeasible (X).
- 5. Which risks have NO feasible countermeasure? These are your biggest vulnerabilities.
- 6. Present to the class (3 min per team).
- Time: 20 minutes.



Part III: Dashboards, Scorecards & Spider Charts

Communicating Quality Performance to Stakeholders

Spider Charts (Radar Charts)

- Present multiple metrics simultaneously in a two-dimensional plane.
- Each axis represents a different performance metric; values plotted and connected.
- SCM use: Compare supplier performance across multiple dimensions at a glance.
- Example axes: Quality, Cost, Delivery, Flexibility, Innovation, Sustainability.
- Strengths: Easy visual comparison of multi-dimensional performance.
- Limitations: Hard to read with too many dimensions; scale must be consistent.
- Best practice: Use 5–8 dimensions; normalize scores to a common scale (e.g., 1–5).

Balanced Scorecards for Supply Chain Quality

- A strategic management tool with four perspectives:
 - 1. Financial – Cost of quality, COPQ, savings from improvements.
 - 2. Customer – On-time delivery, defect rates, customer satisfaction.
 - 3. Internal Processes – Cycle time, process capability, first-pass yield.
 - 4. Learning & Growth – Training hours, certifications, innovation metrics.
- Reported regularly: weekly, monthly, quarterly, annually.
- In SCM: Extend the scorecard to include supplier-specific metrics.
- Each supplier can have their own balanced scorecard, reviewed quarterly.
- Links quality performance to strategic goals – not just ‘firefighting.’

Dashboards: Real-Time Quality Visibility

- Visual displays that communicate performance levels quickly and clearly.
- Design principles: Simple, clear, actionable – like a car dashboard.
- Common elements: Gauges (red/yellow/green), trend lines, KPI cards, alerts.
- Modern SCM dashboards: Real-time data from IoT sensors, ERP systems, supplier portals.
- Examples of SCM dashboard KPIs:
 - – Supplier quality index (SQI) by supplier and part number.
 - – On-time delivery rate with trend.
 - – Cost of poor quality (COPQ) by category.
 - – Customer complaint rate with Pareto breakdown.
- Key: A dashboard should answer ‘What do I need to act on NOW?’ in under 5 seconds.

Exercise: Design a Supply Chain Quality Dashboard

Scenario

- You are the VP of Supply Chain Quality for an electronics company.
- You need a one-page dashboard for the weekly executive meeting.
- Your supply chain has 25 suppliers, 3 factories, and 200+ products.
- The CEO wants to know: Are we improving? Where are the problems?

Tasks

- 1. Select 6–8 KPIs for your dashboard. Justify each choice.
- 2. Sketch the dashboard layout on paper (or use PowerPoint/Excel).
- 3. Decide: Which KPIs get gauges? Trend lines? Tables?
- 4. What color-coding rules would you use (red/yellow/green)?
- 5. How would you drill down from the dashboard to root causes?
- 6. Present your dashboard to the class (3 min).
- Time: 25 minutes.



Capstone Activity

Integrating All Quality Tools in a Supply Chain Scenario

Capstone Simulation: Quality Crisis at TechPro Electronics

The Scenario

- TechPro Electronics manufactures smartphones. In the last quarter:
 - Customer returns increased 85%.
 - 3 major retailers threatened to delist the product.
 - Social media complaints went viral.
- You are the newly appointed Quality Task Force.
- You have data: returns log, supplier list, inspection records, customer complaints.

Your Mission (60 min)

- Phase 1 (10 min): Use Check Sheet data to create a Pareto Chart of defect types.
- Phase 2 (10 min): Build a Fishbone Diagram for the #1 defect category.
- Phase 3 (10 min): Use a Prioritization Matrix to rank corrective actions.
- Phase 4 (10 min): Create a PDPC for implementing the top corrective action.
- Phase 5 (10 min): Design a dashboard to monitor improvement.
- Phase 6 (10 min): Present your complete analysis to 'the Board' (class).

Capstone Data Pack (Provided to Each Team)

Returns Data (500 units returned)

- Defect Type | Count
- Screen cracking | 185
- Battery swelling | 120
- Camera malfunction | 78
- Speaker failure | 52
- Charging port loose | 35
- Software glitch | 18
- Other | 12
-
- Top 3 suppliers by defect contribution:
- ScreenTech (screens): 160 defects
- PowerCell (batteries): 115 defects
- OptiLens (cameras): 72 defects

Additional Information

- ScreenTech changed glass supplier 3 months ago.
- PowerCell is running at 95% capacity utilization.
- OptiLens had 2 key engineers resign last quarter.
- Your factory's incoming inspection catches only 30% of defects.
- No control charts are currently in use for incoming inspection.
- Supplier audits are conducted only annually.
-
- Use this data to drive your analysis through all 6 phases.
- Be prepared to justify every recommendation with data and tools.

Summary & Reflection

Connecting the Tools to Supply Chain Excellence

Summary: Quality Tools Toolkit for SCM

- B7 Tools (Operational): Process Maps, Check Sheets, Histograms, Scatter Plots,
 - Control Charts, Fishbone Diagrams, Pareto Charts.
 - → Use when you have DATA and need to analyze, visualize, and solve specific problems.
 -
- N7 Tools (Managerial): Affinity Diagram, Interrelationship Digraph, Tree Diagram,
 - Prioritization Matrix, Matrix Diagram, PDPC, Activity Network Diagram.
 - → Use when you have IDEAS and need to organize, prioritize, and plan.
 -
- Reporting Tools: Spider Charts, Balanced Scorecards, Dashboards.
 - → Use when you need to COMMUNICATE performance and drive accountability.
 -
- The power is in COMBINING tools: Check Sheet → Pareto → Fishbone → Control Chart → Dashboard.

Final Reflection Questions

- 1. Which quality tool do you think is the MOST underused in supply chain management? Why?
- 2. How would you convince a skeptical supplier to adopt SPC (control charts)?
- 3. Can these tools work in a services supply chain (e.g., healthcare, logistics)? How would they differ?
- 4. With Industry 4.0 and IoT, how are these traditional tools evolving?
 - – Real-time dashboards replace periodic check sheets?
 - – AI-powered fishbone diagrams?
 - – Automated Pareto analysis from ERP data?
- 5. If you could only use THREE quality tools for the rest of your career, which three would you pick and why?

Key Takeaways

- Quality tools are not academic exercises – they are daily-use instruments for SCM professionals.
- Start simple: Process Map → Check Sheet → Pareto. These three alone can solve 60% of problems.
- Data-driven decisions beat gut feeling every time. Let the data tell the story.
- In supply chains, quality is a SHARED responsibility. Tools must cross organizational boundaries.
- The best quality tool is the one you actually USE. Master a few, then expand your toolkit.
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- "Quality is never an accident; it is always the result of intelligent effort." – John Ruskin

Quick Reference: Which Tool When?

If you need to...

- Visualize a process → Process Map / VSM
- Collect data systematically → Check Sheet
- See data distribution → Histogram
- Find variable relationships → Scatter Plot
- Monitor process stability → Control Chart
- Find root causes → Fishbone Diagram
- Prioritize problems → Pareto Chart

If you need to...

- Organize brainstorming → Affinity Diagram
- Find root drivers → Interrelationship Digraph
- Break down goals → Tree Diagram
- Rank alternatives → Prioritization Matrix
- Show multi-dimensional links → Matrix Diagram
- Plan for contingencies → PDPC
- Schedule a project → Activity Network
- Communicate performance → Dashboard / Scorecard / Spider Chart